

EXPERIMENT - 07

Question 1

Write a C program to find the painting cost of a cube when side and painting rate per square meter are given.

Step 1: Understand the Problem Statement

Problem: The cube has 6 equal square faces. We need to compute the total surface area of the cube and then find the cost of painting that area at a given rate per square meter.

Required Input: side of the cube (in meters), painting rate (cost per square meter)

Required Output: total surface area of the cube and the total painting cost

Formula Used:

- Surface Area of Cube = $6 \times \text{side} \times \text{side}$
- Painting Cost = Surface Area $\times$ Rate per square meter

Step 2: Prepare the Test Case Table

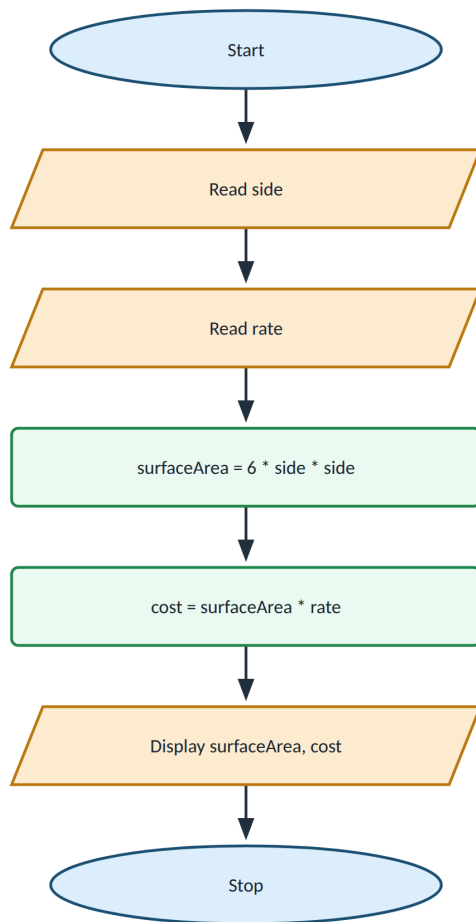
Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	side = 5 rate = 10	Surface Area = 150.00 sq.m Cost = 1500.00	Surface Area = 150.00 sq.m Cost = 1500.00	Pass
2	side = 2.5 rate = 15	Surface Area = 37.50 sq.m Cost = 562.50	Surface Area = 37.50 sq.m Cost = 562.50	Pass
3	side = 1 rate = 100	Surface Area = 6.00 sq.m Cost = 600.00	Surface Area = 6.00 sq.m Cost = 600.00	Pass
4	side = 0 rate = 20	Surface Area = 0.00 sq.m Cost = 0.00	Surface Area = 0.00 sq.m Cost = 0.00	Pass

Step 3: Algorithm

1. Start
2. Read the value of side of the cube
3. Read the value of painting rate per square meter
4. Compute $\text{surfaceArea} = 6 * \text{side} * \text{side}$
5. Compute $\text{cost} = \text{surfaceArea} * \text{rate}$
6. Display surfaceArea and cost
7. Stop

Evaluation against test cases: Tracing the algorithm manually for each test case (e.g., for side = 5 and rate = 10, surfaceArea = $6 \times 5 \times 5 = 150$ and cost = $150 \times 10 = 1500$) produces results that exactly match the expected output for all 4 test cases listed in Step 2. Hence the algorithm is logically correct.

Step 4: Flowchart



Evaluation against test cases: The flowchart follows the same sequence as the algorithm — read side, read rate, compute surfaceArea, compute cost, then display the results. Walking through each of the 4 test cases along this flow produces the same values obtained in Step 3, confirming the flowchart correctly represents the logic.

Step 5: C Program

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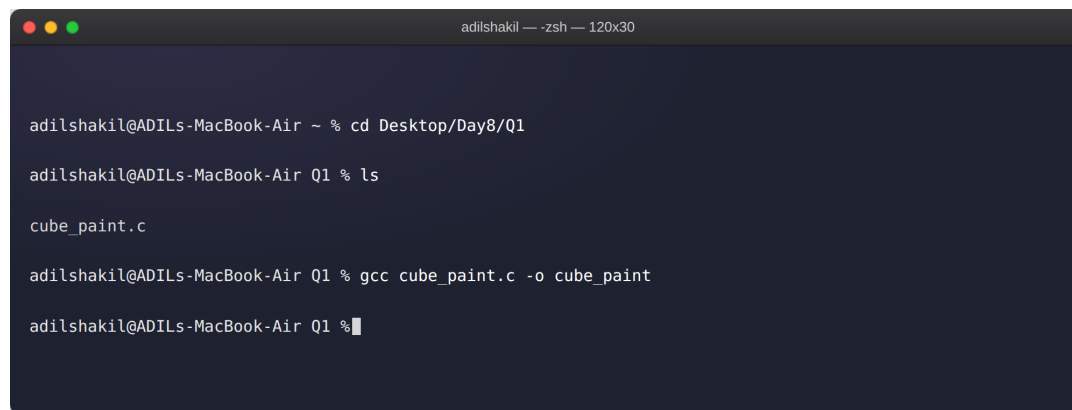
Step 6: Compile, Execute, and Evaluate

The program was compiled using the GCC compiler on macOS Terminal (zsh) with the command shown below. The compilation completed with no errors or warnings, and the executable was run once for each of the 4 test cases prepared in Step 2.

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	side = 5 rate = 10	Surface Area = 150.00 sq.m Cost = 1500.00	Surface Area = 150.00 sq.m Cost = 1500.00	Pass
2	side = 2.5 rate = 15	Surface Area = 37.50 sq.m Cost = 562.50	Surface Area = 37.50 sq.m Cost = 562.50	Pass
3	side = 1 rate = 100	Surface Area = 6.00 sq.m Cost = 600.00	Surface Area = 6.00 sq.m Cost = 600.00	Pass
4	side = 0 rate = 20	Surface Area = 0.00 sq.m Cost = 0.00	Surface Area = 0.00 sq.m Cost = 0.00	Pass

Actual output for all 4 test cases matches the expected output exactly, so every test case is marked Pass.

Step 7: Compilation and Execution Screenshots



```
adilshakil@ADILs-MacBook-Air ~ % cd Desktop/Day8/Q1
adilshakil@ADILs-MacBook-Air Q1 % ls
cube_paint.c
adilshakil@ADILs-MacBook-Air Q1 % gcc cube_paint.c -o cube_paint
adilshakil@ADILs-MacBook-Air Q1 %
```

```
adilshakil@ADILs-MacBook-Air Q1 % ./cube_paint
```

```
Enter the side of the cube (in meters): 5
```

```
Enter the painting rate (per square meter): 10
```

```
Surface Area of the cube = 150.00 sq. meters
```

```
Total Painting Cost = 1500.00
```

```
adilshakil@ADILs-MacBook-Air Q1 % ./cube_paint
```

```
Enter the side of the cube (in meters): 2.5
```

```
Enter the painting rate (per square meter): 15
```

```
Surface Area of the cube = 37.50 sq. meters
```

```
Total Painting Cost = 562.50
```

```
adilshakil@ADILs-MacBook-Air Q1 % ./cube_paint
```

```
Enter the side of the cube (in meters): 1
```

```
Enter the painting rate (per square meter): 100
```

```
Surface Area of the cube = 6.00 sq. meters
```

```
Total Painting Cost = 600.00
```

```
adilshakil@ADILs-MacBook-Air Q1 % ./cube_paint
```

```
Enter the side of the cube (in meters): 0
```

```
Enter the painting rate (per square meter): 20
```

```
Surface Area of the cube = 0.00 sq. meters
```

```
Total Painting Cost = 0.00
```

```
adilshakil@ADILs-MacBook-Air Q1 %
```

Question 2

Write a C program to find the length of an arc when radius and central angle are given.

Step 1: Understand the Problem Statement

Problem: An arc is a part of the circumference of a circle. Given the radius of the circle and the central angle (in degrees) subtended by the arc, we need to compute the length of that arc.

Required Input: radius of the circle, central angle in degrees

Required Output: length of the arc

Formula Used:

- Arc Length = $(\text{angle} / 360) \times 2 \times \pi \times \text{radius}$

Step 2: Prepare the Test Case Table

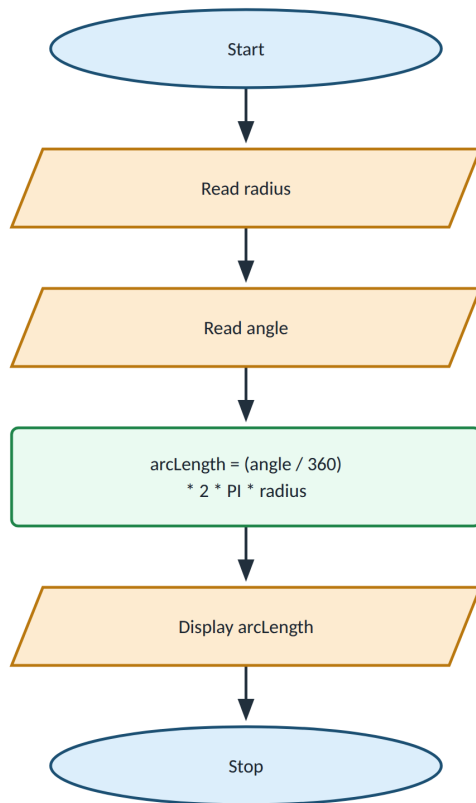
Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	radius = 7 angle = 90	Arc Length = 10.9956 units	Arc Length = 10.9956 units	Pass
2	radius = 10 angle = 180	Arc Length = 31.4159 units	Arc Length = 31.4159 units	Pass
3	radius = 5 angle = 360	Arc Length = 31.4159 units	Arc Length = 31.4159 units	Pass
4	radius = 3 angle = 45	Arc Length = 2.3562 units	Arc Length = 2.3562 units	Pass

Step 3: Algorithm

1. Start
2. Read the value of radius
3. Read the value of the central angle (in degrees)
4. Compute $\text{arcLength} = (\text{angle} / 360) * 2 * \text{PI} * \text{radius}$
5. Display arcLength
6. Stop

Evaluation against test cases: Tracing the algorithm manually (e.g., for radius = 7 and angle = 90, $\text{arcLength} = (90/360) \times 2 \times 3.14159265 \times 7 = 10.9956$) produces results that exactly match the expected output for all 4 test cases listed in Step 2. Hence the algorithm is logically correct.

Step 4: Flowchart



Evaluation against test cases: The flowchart follows the same sequence as the algorithm — read radius, read angle, compute arcLength, then display the result. Walking through each of the 4 test cases along this flow produces the same values obtained in Step 3, confirming the flowchart correctly represents the logic.

Step 5: C Program

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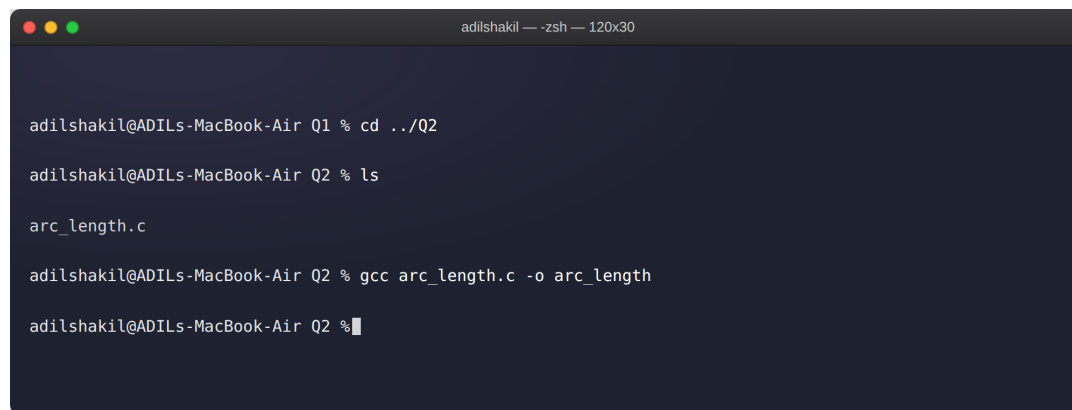
Step 6: Compile, Execute, and Evaluate

The program was compiled using the GCC compiler on macOS Terminal (zsh) with the command shown below. The compilation completed with no errors or warnings, and the executable was run once for each of the 4 test cases prepared in Step 2.

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	radius = 7 angle = 90	Arc Length = 10.9956 units	Arc Length = 10.9956 units	Pass
2	radius = 10 angle = 180	Arc Length = 31.4159 units	Arc Length = 31.4159 units	Pass
3	radius = 5 angle = 360	Arc Length = 31.4159 units	Arc Length = 31.4159 units	Pass
4	radius = 3 angle = 45	Arc Length = 2.3562 units	Arc Length = 2.3562 units	Pass

Actual output for all 4 test cases matches the expected output exactly, so every test case is marked Pass.

Step 7: Compilation and Execution Screenshots



```
adilshakil@ADILs-MacBook-Air Q1 % cd ../Q2
adilshakil@ADILs-MacBook-Air Q2 % ls
arc_length.c
adilshakil@ADILs-MacBook-Air Q2 % gcc arc_length.c -o arc_length
adilshakil@ADILs-MacBook-Air Q2 %
```

```
adilshakil@ADILs-MacBook-Air Q2 % ./arc_length
```

```
Enter the radius of the circle: 7
```

```
Enter the central angle (in degrees): 90
```

```
Length of the Arc = 10.9956 units
```

```
adilshakil@ADILs-MacBook-Air Q2 % ./arc_length
```

```
Enter the radius of the circle: 10
```

```
Enter the central angle (in degrees): 180
```

```
Length of the Arc = 31.4159 units
```

```
adilshakil@ADILs-MacBook-Air Q2 % ./arc_length
```

```
Enter the radius of the circle: 5
```

```
Enter the central angle (in degrees): 360
```

```
Length of the Arc = 31.4159 units
```

```
adilshakil@ADILs-MacBook-Air Q2 % ./arc_length
```

```
Enter the radius of the circle: 3
```

```
Enter the central angle (in degrees): 45
```

```
Length of the Arc = 2.3562 units
```

```
adilshakil@ADILs-MacBook-Air Q2 %
```